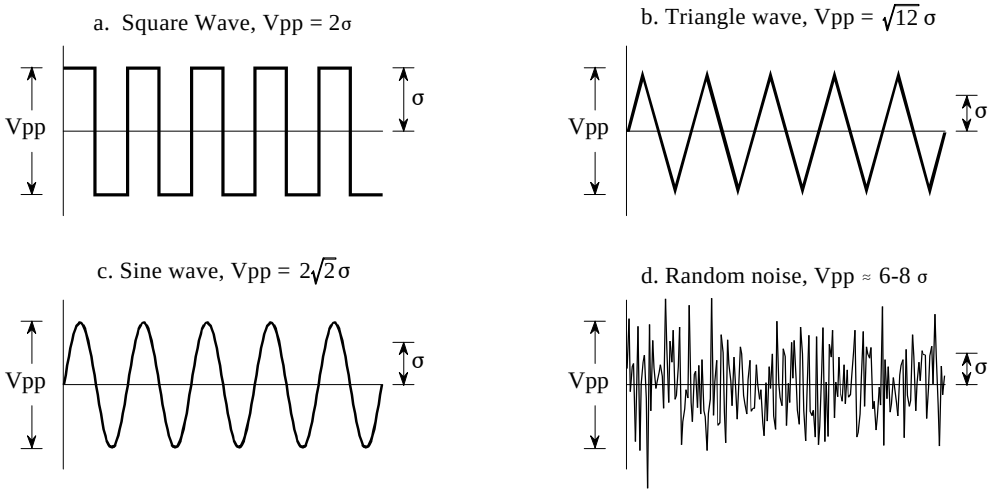

FIGURE 2-2
Ratio of the peak-to-peak amplitude to the standard deviation for several common waveforms. For the square wave, this ratio is 2; for the triangle wave it is $\sqrt{12} = 3.46$; for the sine wave it is $2\sqrt{2} = 2.83$. While random noise has no *exact* peak-to-peak value, it is *approximately* 6 to 8 times the standard deviation.

Table 2-1 lists a computer routine for calculating the mean and standard deviation using Eqs. 2-1 and 2-2. The programs in this book are intended to convey *algorithms* in the most straightforward way; all other factors are treated as secondary. Good programming techniques are disregarded if it makes the program logic more clear. For instance: a simplified version of BASIC is used, line numbers are included, the only control structure allowed is the FOR-NEXT loop, there are no I/O statements, etc. Think of these programs as an alternative way of understanding the equations used

```
100 CALCULATION OF THE MEAN AND STANDARD DEVIATION
110 '
120 DIM X[511]                    'The signal is held in X[0] to X[511]
130 N% = 512                      'N% is the number of points in the signal
140 '
150 GOSUB XXXX                    'Mythical subroutine that loads the signal into X[ ]
160 '
170 MEAN = 0                      'Find the mean via Eq. 2-1
180 FOR I% = 0 TO N%-1
190   MEAN = MEAN + X[I%]
200 NEXT I%
210 MEAN = MEAN/N%
220 '
230 VARIANCE = 0                  'Find the standard deviation via Eq. 2-2
240 FOR I% = 0 TO N%-1
250   VARIANCE = VARIANCE + ( X[I%] - MEAN )^2
260 NEXT I%
270 VARIANCE = VARIANCE/(N%-1)
280 SD = SQR(VARIANCE)
290 '
300 PRINT  MEAN  SD               'Print the calculated mean and standard deviation
310 '
320 END
```

TABLE 2-1